

PHOTOTRANSDUCTION

VISUAL CYCLE

- Rods contains a photosensitive pigment known as rhodopsin.
- Rhodopsin is also called as visual purple.
- It is a combination of scotopsin and retinal.
- Retinal here is 11-cis retinal.
- Exposure to light causes isomerization of 11-cis retinal into all-trans retinal
- It has a peak sensitivity to light @ a wavelength of 505nm.

Photoreceptors

- Each eye has 130 million rods for Scotopic/night vision, 5.5 million cones for Photopic/bright light and color vision
- PR contains photopigment that absorbs light and becomes active
- It has 2 components:

Opsin: integral protein of disc membrane

Retinal: vitamin A derivative binds interior of opsin (light absorbing part)

- **Rods:** Rhodopsin (retinene + scotopsin)
- **Opsin in 3 cones called Photopsin 1. Prophyropsin 2. Iodopsin 3. Cyanopsin**
- Red cones respond to longest & blue cones to shortest wavelengths of light

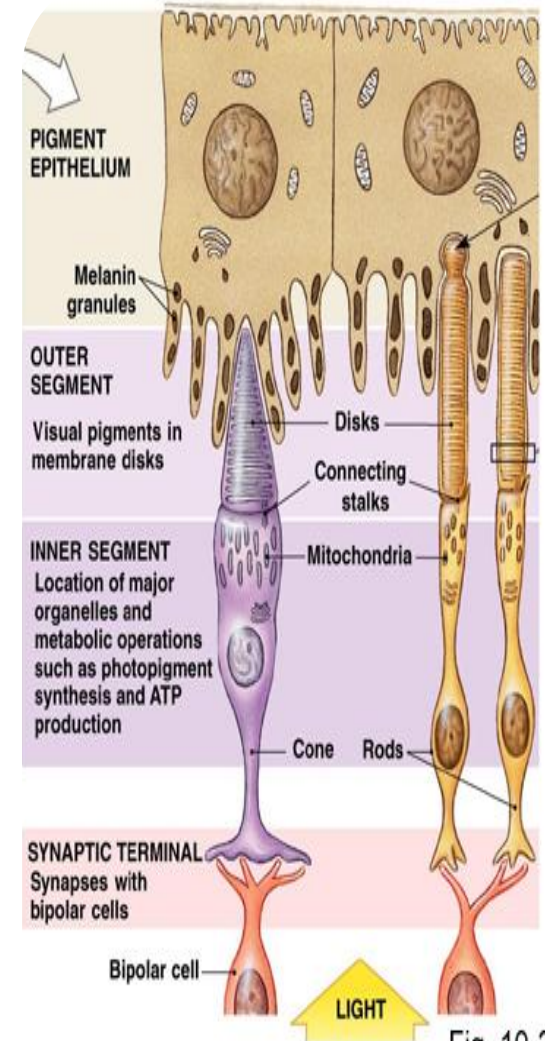


Fig. 10-37

PHOTOTRANSDUCTION

- Visual or phototransduction is the process by which light energy is converted into receptor potential in visual receptors.
- Resting membrane potential in other sensory receptor cells is usually between -70 and -90 mV. However, in the visual receptors during darkness, negativity is reduced and resting membrane potential is about -40 mV. It is because of influx of sodium ions.
- Normally in dark, sodium ions are pumped out of inner segments of rod cell to ECF. However, these sodium ions leak back into the rod cells through membrane of outer segment and reduce the electronegativity inside rod cell. Thus, sodium influx maintains a decreased negative potential up to -40 mV.

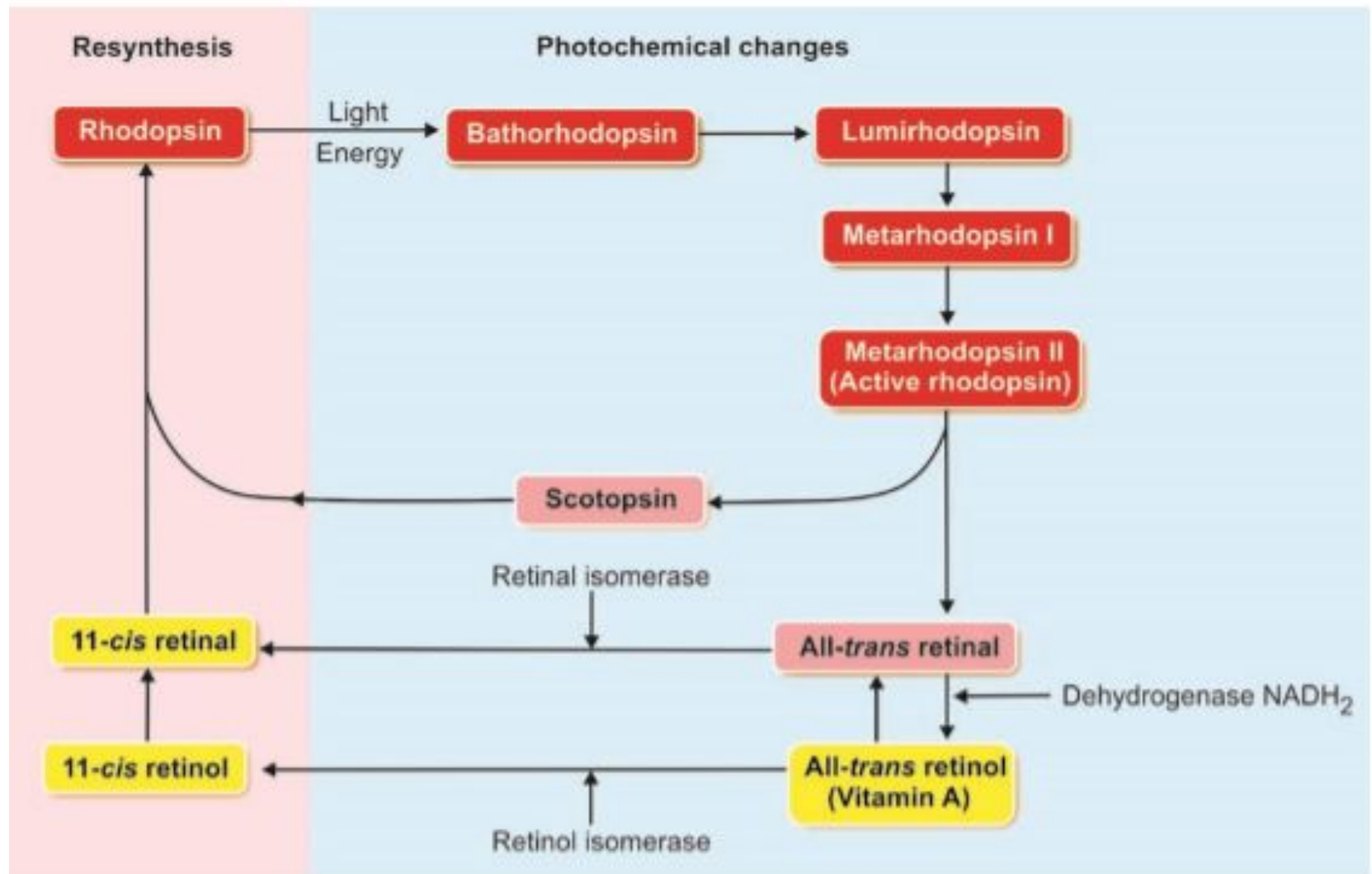


FIGURE 166.2: Photochemical changes and resynthesis of rhodopsin (Wald visual cycle).
NADH₂ = Reduced nicotinamide adenine dinucleotide.

Influx of sodium ions into outer segment of rod cell occurs mainly because of cyclic guanosine monophosphate (cGMP) present in the cytoplasm of cell. The cGMP always keeps the sodium channels opened. Closure of sodium channels occurs due to reduction in cGMP.

Concentration of sodium ions inside the rod cell is regulated by sodium potassium pump. When light falls on retina, rhodopsin is excited leading to development of receptor potential in the rod cells.

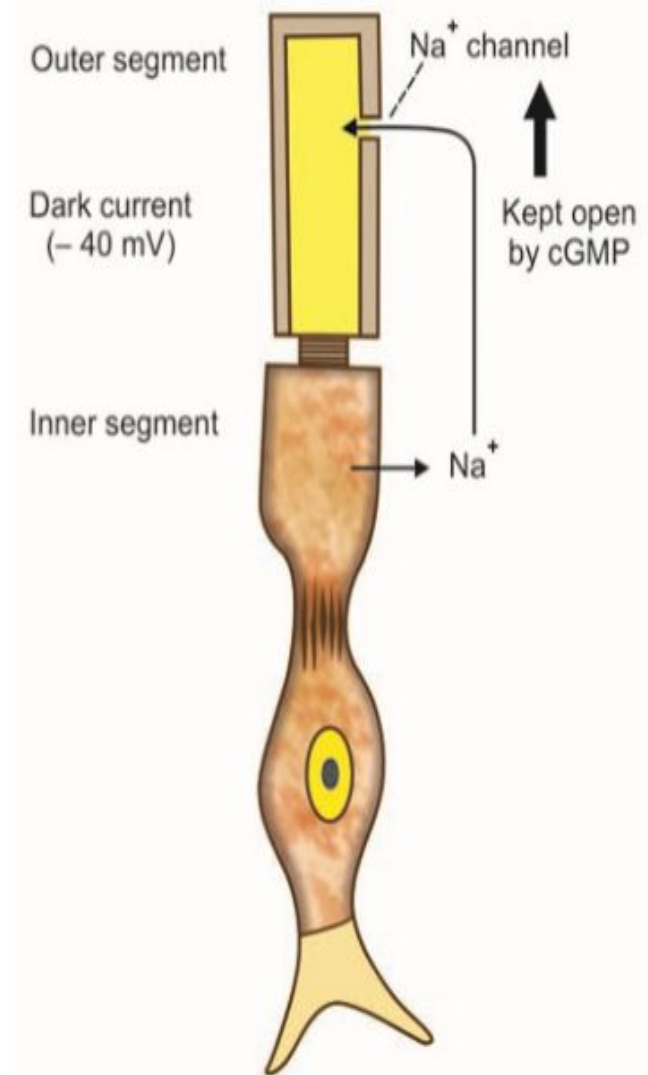


FIGURE 166.3: Maintenance of dark current (resting potential) in outer segment of rod cell

Photo transduction Cascade of Receptor Potential

- Following is the phototransduction cascade of receptor potential :
 1. When a photon (the minimum quantum of light energy) is absorbed by rhodopsin, the 11-cis retinal is decomposed into metarhodopsin through few reactions. Metarhodopsin II is considered as the active form of rhodopsin. It plays an important role in the development of receptor potential.
 2. Metarhodopsin II activates a G protein called transducin that is present in rod disks.
 3. Activated transducin activates the enzyme called cyclic guanosine monophosphate phospho diesterase (cGMP phosphodiesterase), which is also present in rod disks.

4. Activated cGMP phosphodiesterase hydrolyzes cGMP to 5'-GMP

5. Now, the concentration of cGMP is reduced in rod cell

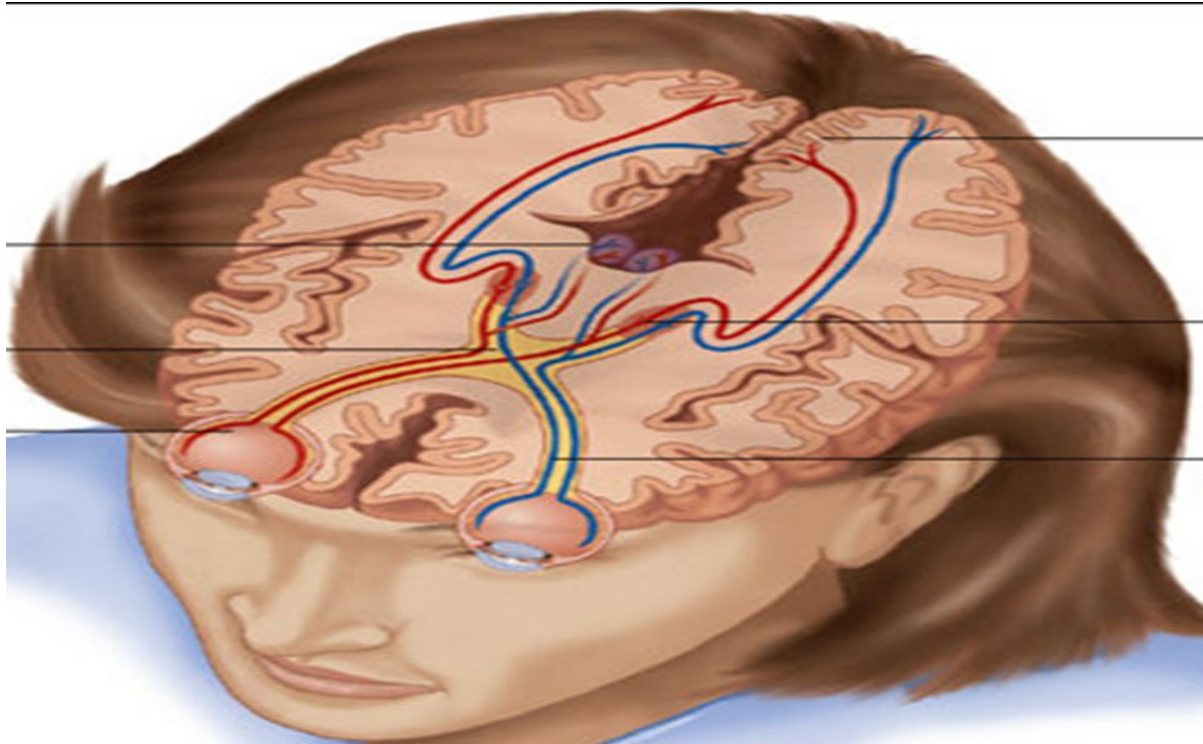
6. Reduction in concentration of cGMP immediately causes closure of sodium channels in the membrane of visual receptors

7. Sudden closure of sodium channels prevents entry of sodium ions leading to hyperpolarization. The potential reaches -70 to -80 mV. It is because of sodium-potassium pump. Thus, the process of receptor potential in visual receptors is unique in nature. When other sensory receptors are excited, the electrical response is in the form of depolarization (receptor potential). But, in visual receptors, the response is in the form of hyperpolarization

Significance of Hyperpolarization

Hyperpolarization in visual receptor cells reduces the release of synaptic transmitter glutamate. It leads to development of response in bipolar cells and ganglionic cells so that, the action potentials are transmitted to cerebral cortex via optic pathway

“VISUAL PATHWAY AND ITS LESIONS”



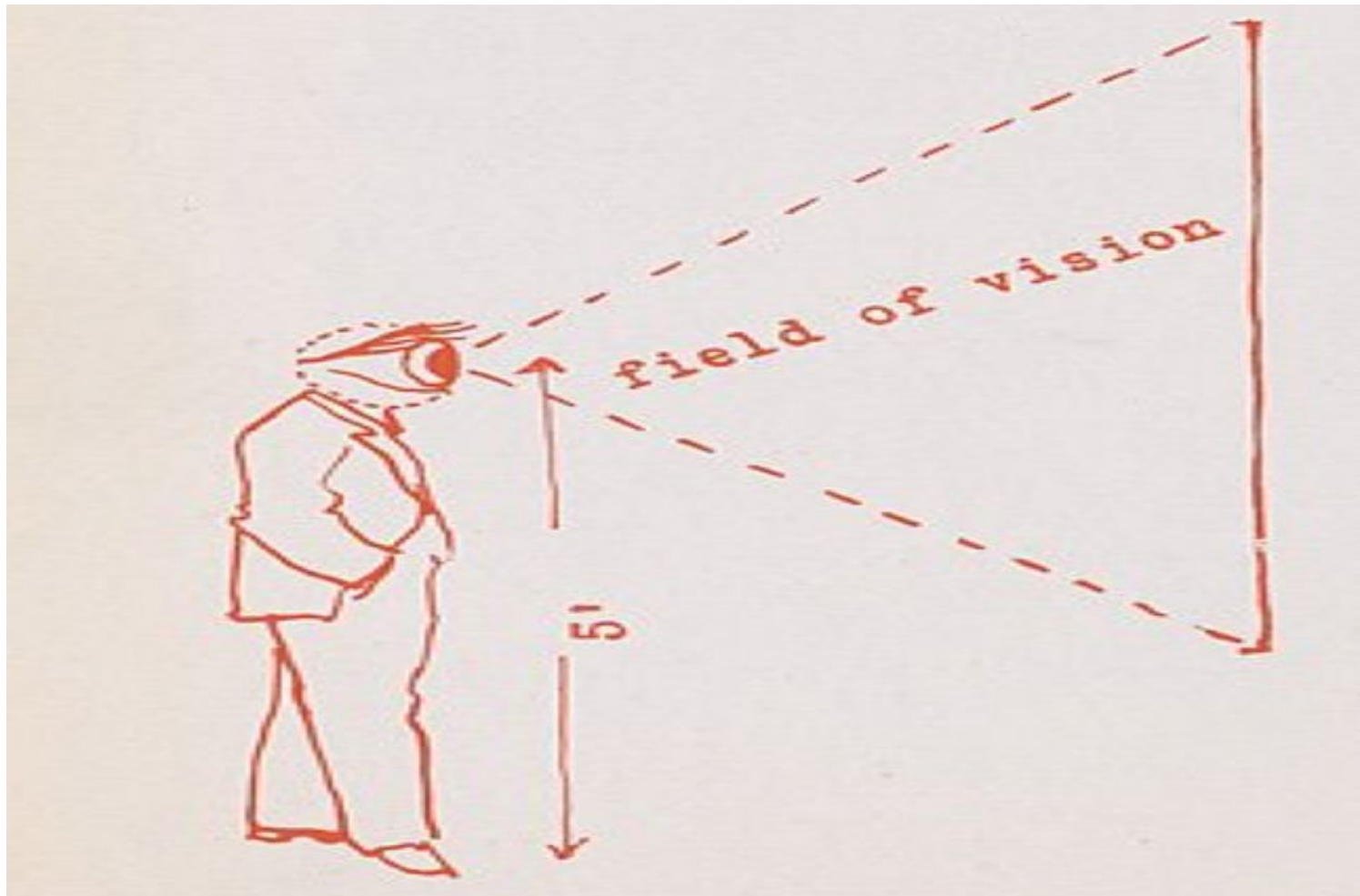
LEARNING OBJECTIVES

At the end of lecture, students are able to

- Discuss the formation of visual pathway
- Describe the structure and function of lateral geniculate body
- Discuss the function of G and M ganglionic cells
- Understand different visual defects

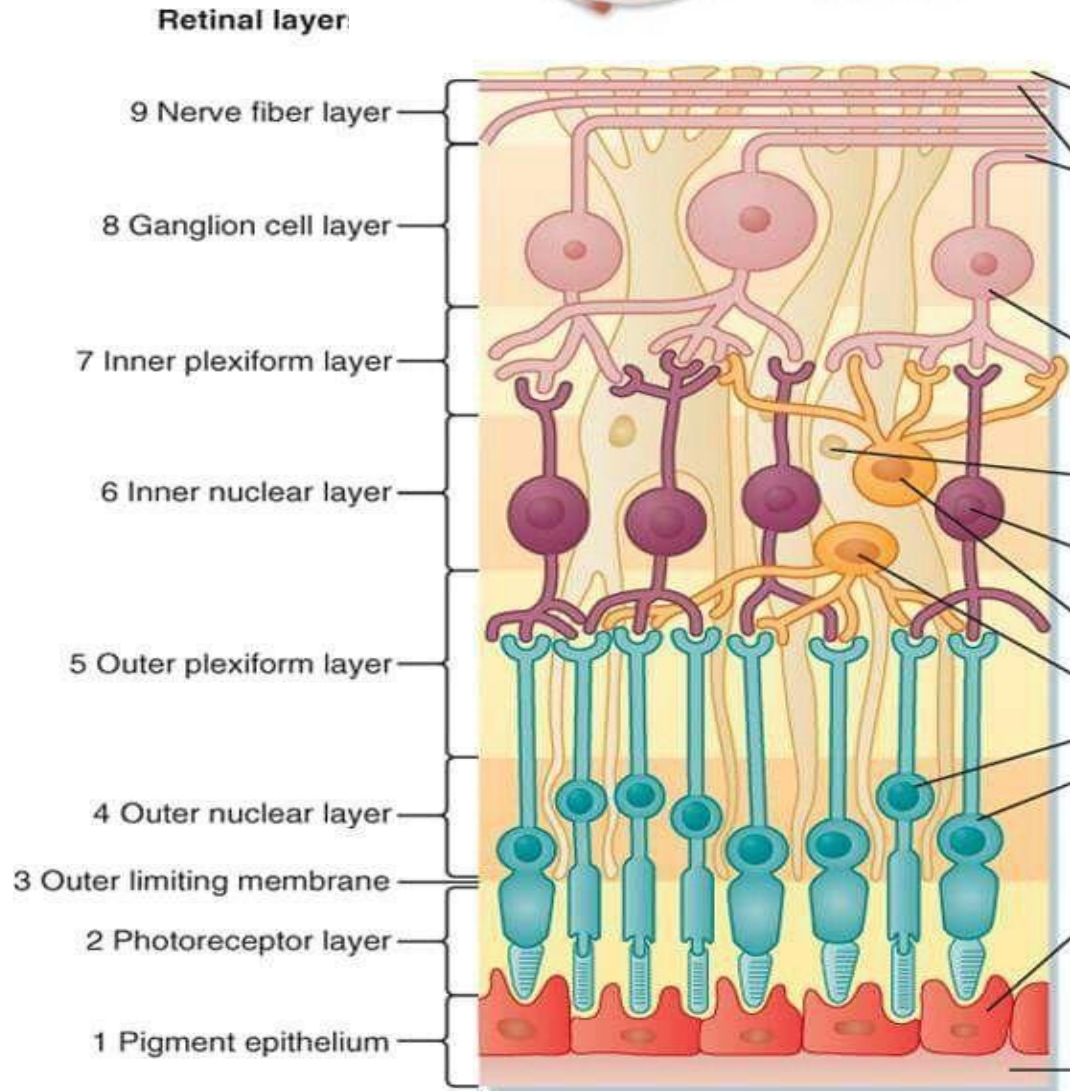
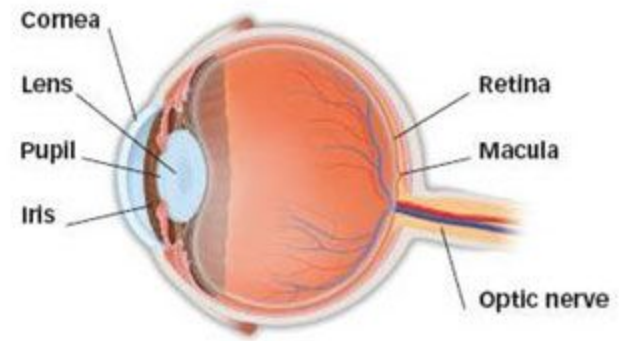
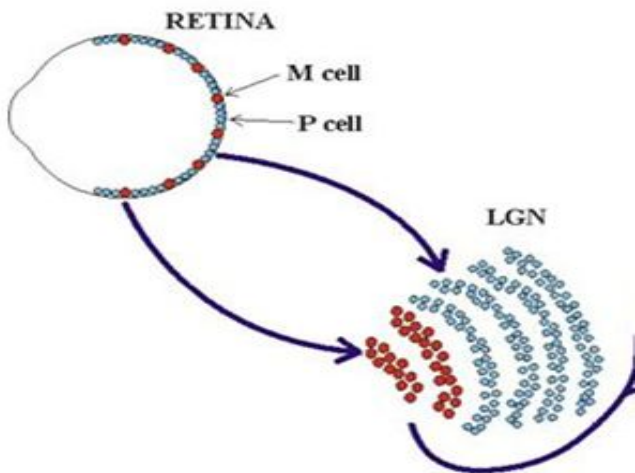
Field of Vision

How far you are able to see without moving your eyes or head.

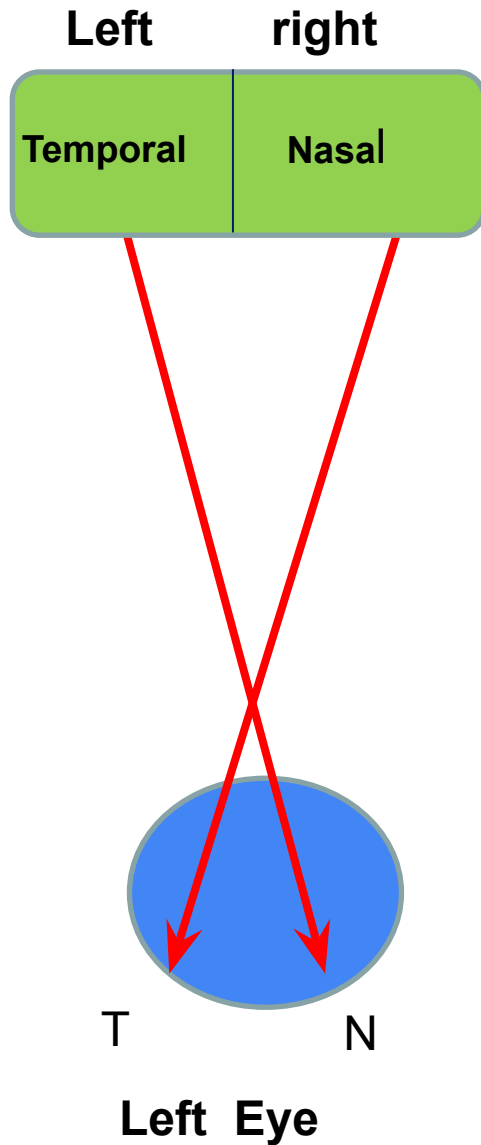


Retinal layers

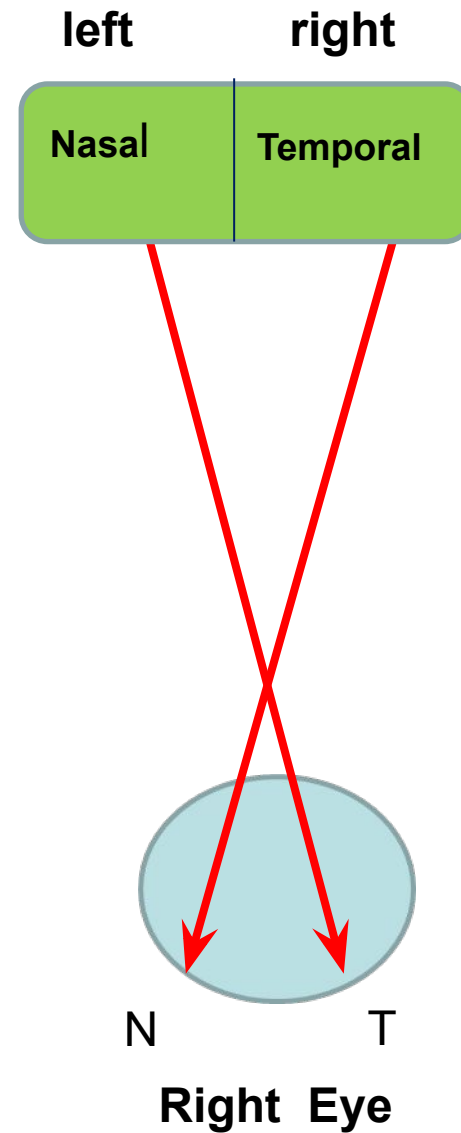
- Axons of ganglionic cells form optic nerve.
- Two important Ganglionic cells are M and P types.



Left Field of vision

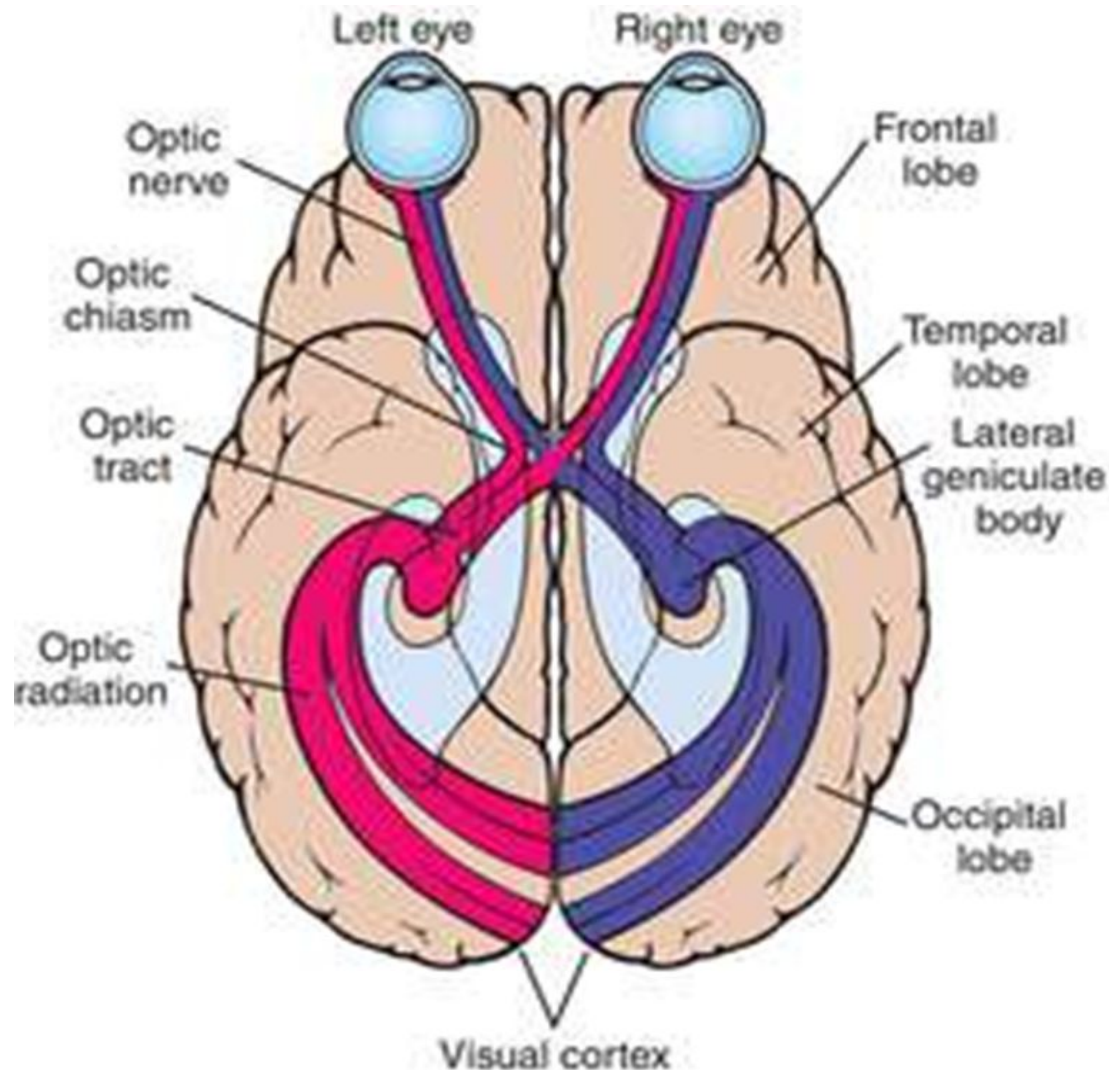


Right Field of vision

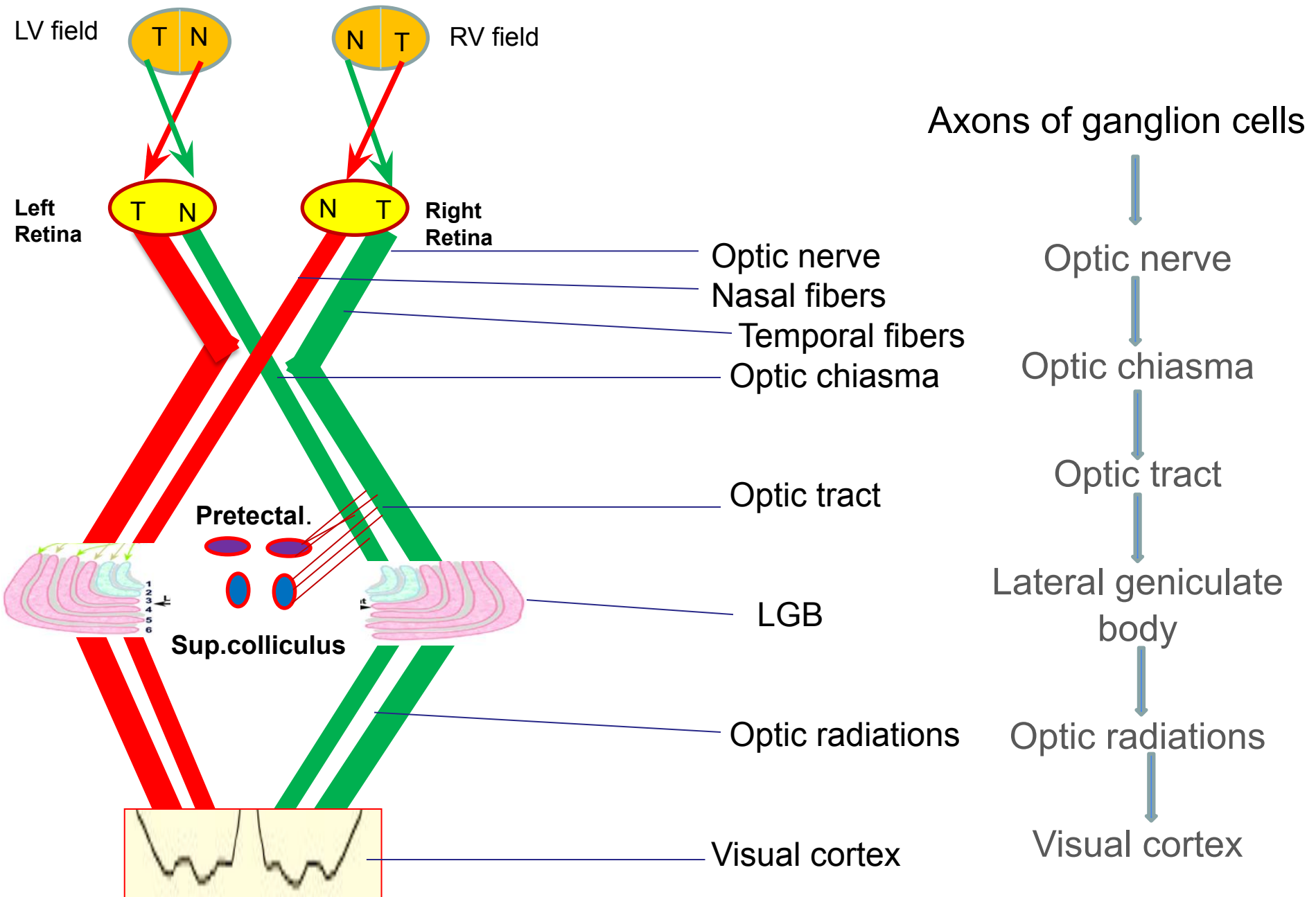


THE PRINCIPAL VISUAL PATHWAYS

(from the two retinas to the visual cortex)



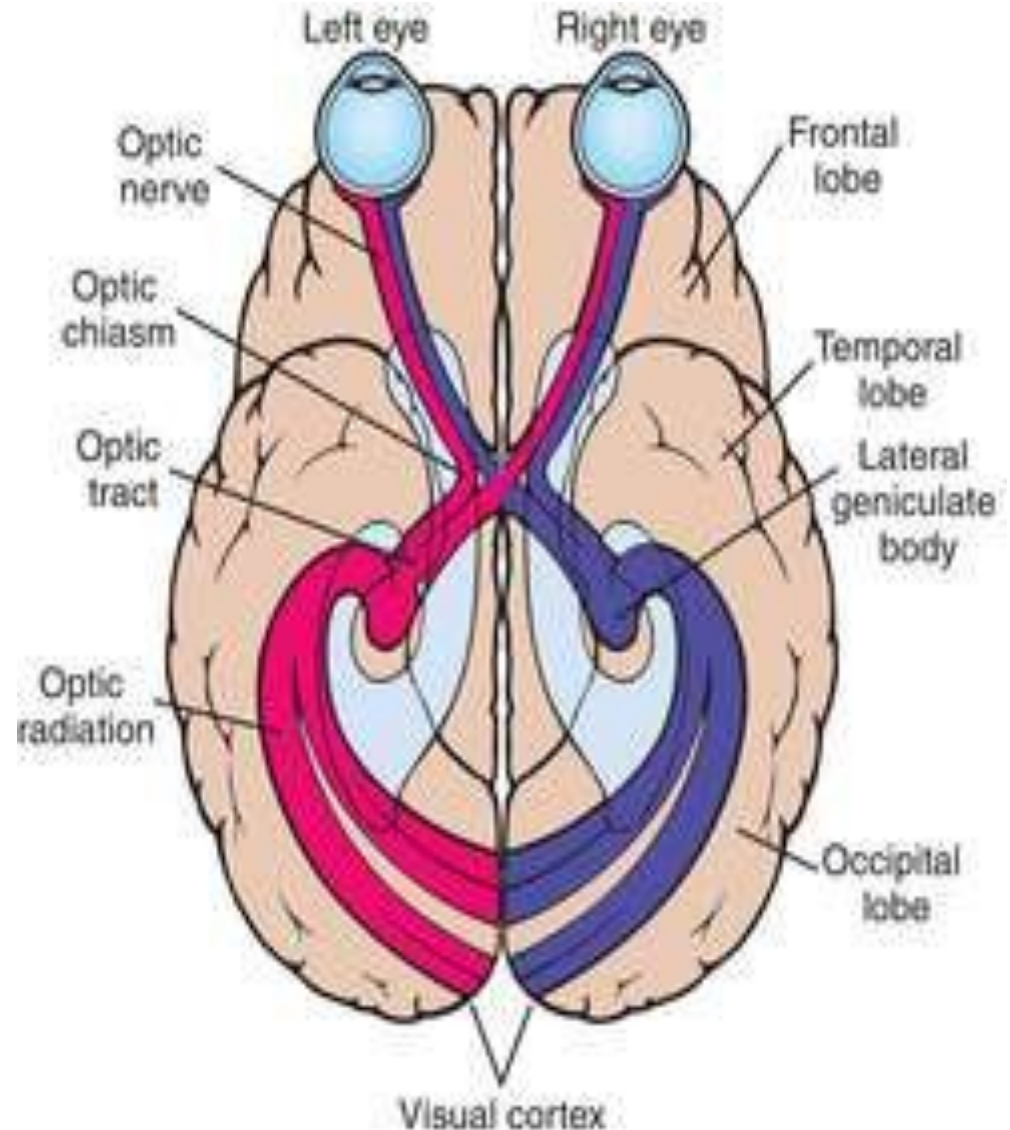
Making of Visual pathway



THE PRINCIPAL VISUAL PATHWAYS

(from the two retinas to the visual cortex)

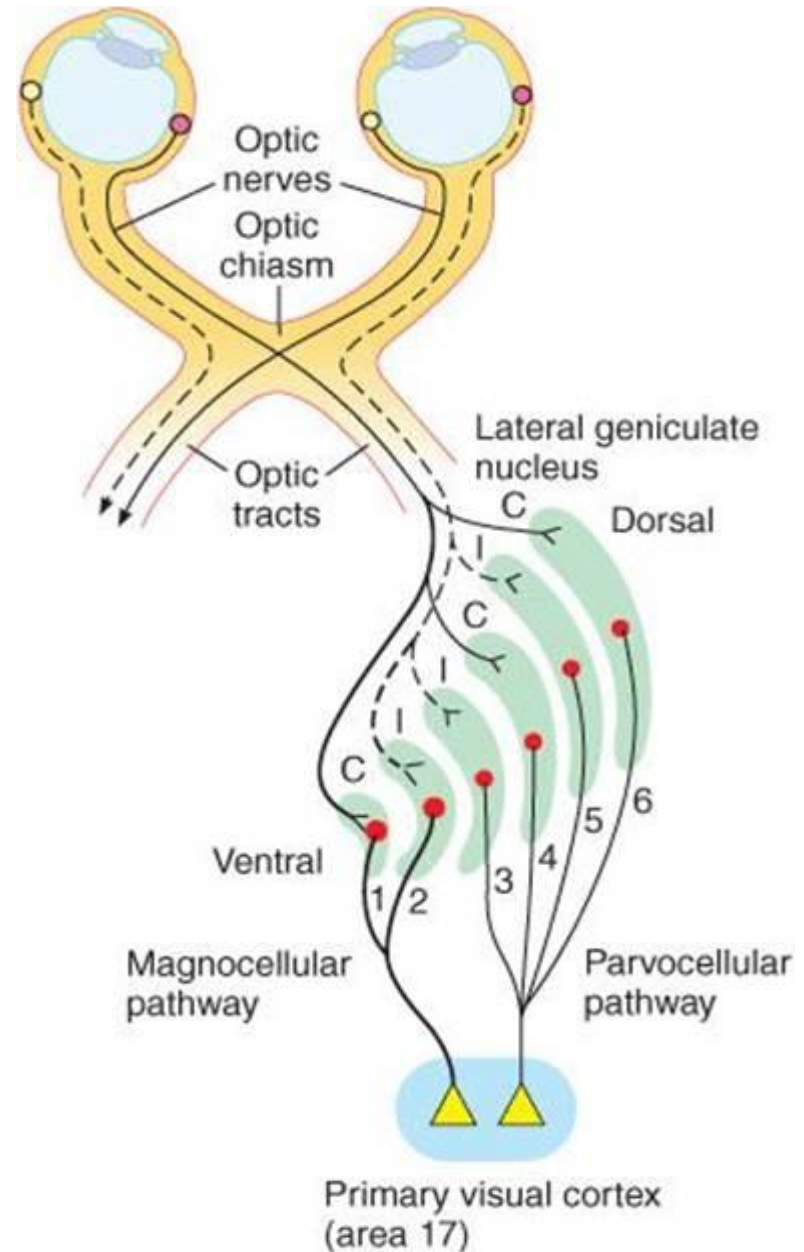
- The visual nerve signals leave the retinas through the optic nerves.
- At the optic chiasma, the optic nerve fibers from the nasal halves of the retinas cross to the opposite sides, where they join the fibers from the opposite temporal retinas to form the optic tracts.
- The fibers of each optic tract synapse in the lateral geniculate nucleus of the thalamus.
- From here, optic radiation pass to the primary visual cortex in the calcarine fissure of the medial occipital lobe.



Lateral Geniculate Nucleus

Six nuclear layers

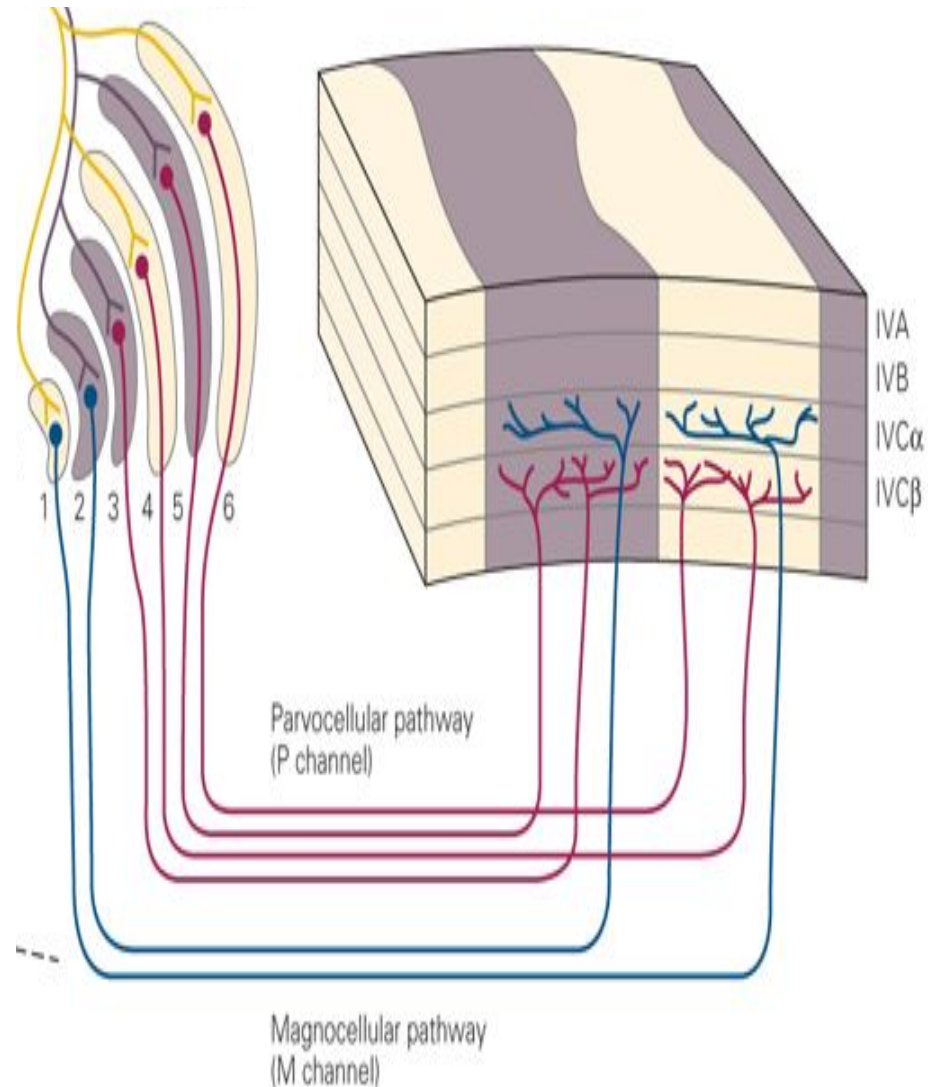
- **Layers II, III and V** receive signals from the ipsilateral retina
- **layers I, IV and VI** receive signals from the contralateral retina.
- **Magnocellular layers**
 - Layers I and II are called magnocellular layers .
 - These neurons receive their input from the large M ganglion cells.
 - This magnocellular system provides a rapidly conducting pathway to the visual cortex
 - Transmitting only black and white information.
 - Its point-to-point transmission is poor
- **Parvocellular layers**
 - Layers III through VI are called parvocellular layers
 - receive their input from the type P retinal ganglion cells
 - only a moderate velocity of conduction
 - Transmit color
 - Convey point-to-point spatial information



Functions of LGB

1. Gates the visual transmission: LGB receives signals from Corticofugal fibers from visual cortex and Reticular areas of mesencephalon to control how much of the signal is allowed to pass to the cortex. They turn off transmission through selected portions of the dorsal LGB.

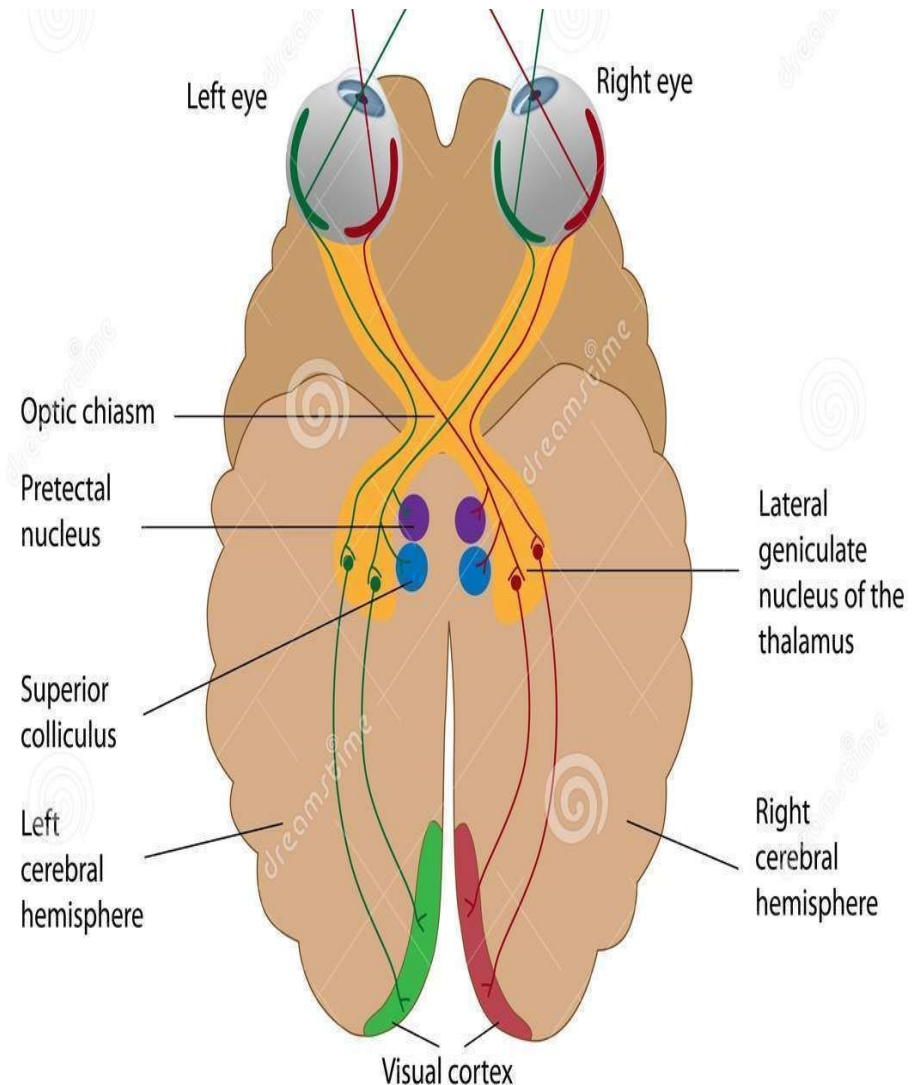
2. Relay function: Point to point visual information from retina to visual cortex.



VISUAL FIBERS TO OTHER AREAS

Visual fibers from optic tract pass to

- Suprachiasmatic nucleus of the hypothalamus, to control circadian rhythms that synchronize physiologic changes of the body with night and day.
- Into the pretectal nuclei in the midbrain, to elicit reflex movements of the eyes to focus on objects of importance and to activate the pupillary light reflex.
- Into the superior colliculus, to control rapid directional movements of the two eyes.



Visual Defects(terminologies)

Anopia

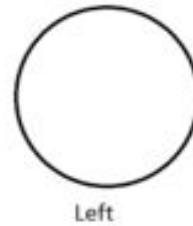
- complete loss of field of vision

Hemianopia

- Loss of half field of vision

Homonymous hemianopia

- Loss of field of vision in same halves of both eyes.
- Types
 - right homonymous hemianopia
 - left homonymous hemianopia



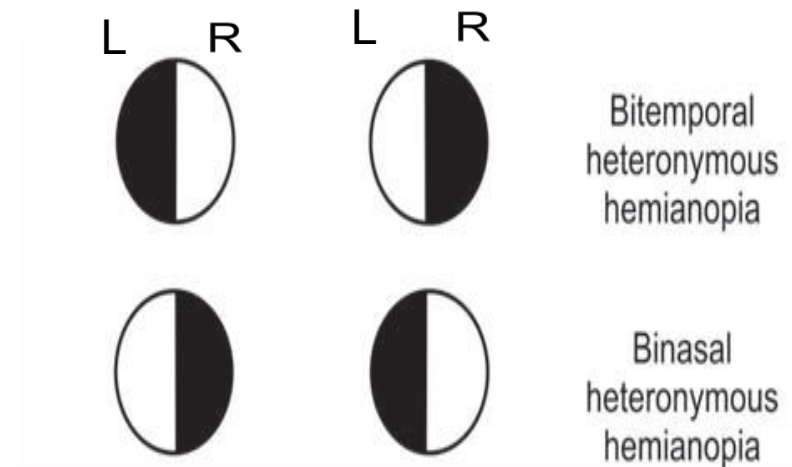
Right
homonymous
hemianopia



Left
homonymous
hemianopia

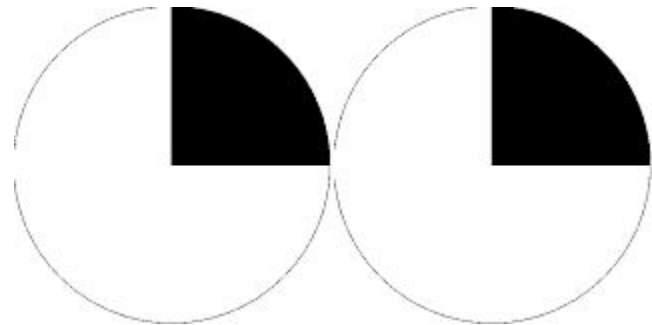
Heteronymous hemianopia

- Loss of field of vision in different halves of both eyes
- Types
 - Bi-temporal heteronymous hemianopia
 - Bi-nasal heteronymous hemianopia



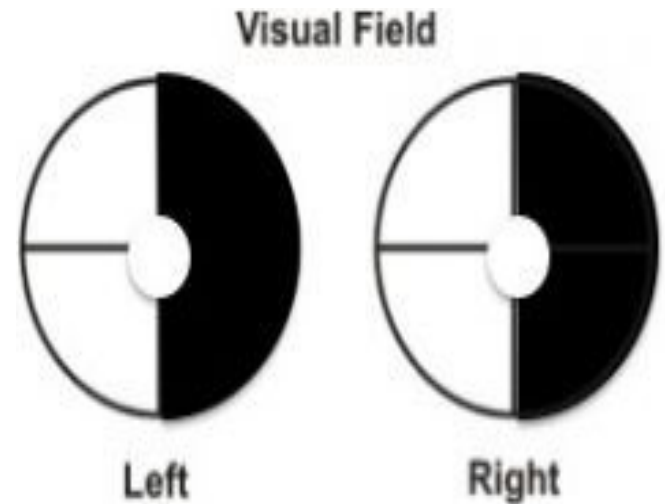
Quadrant anopia

- Loss of field of vision in same quadrant (1/4) of both eyes



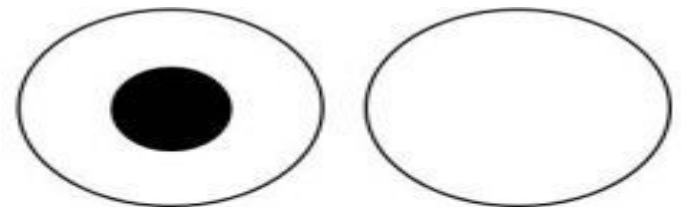
Macular sparing

- loss of peripheral vision with intact macular vision, is also common with occipital lesions
- Because the macular representation is separate from that of the peripheral fields and very large relative to that of the peripheral fields.
- Therefore, occipital lesions must extend considerable distances to destroy macular as well as peripheral vision.

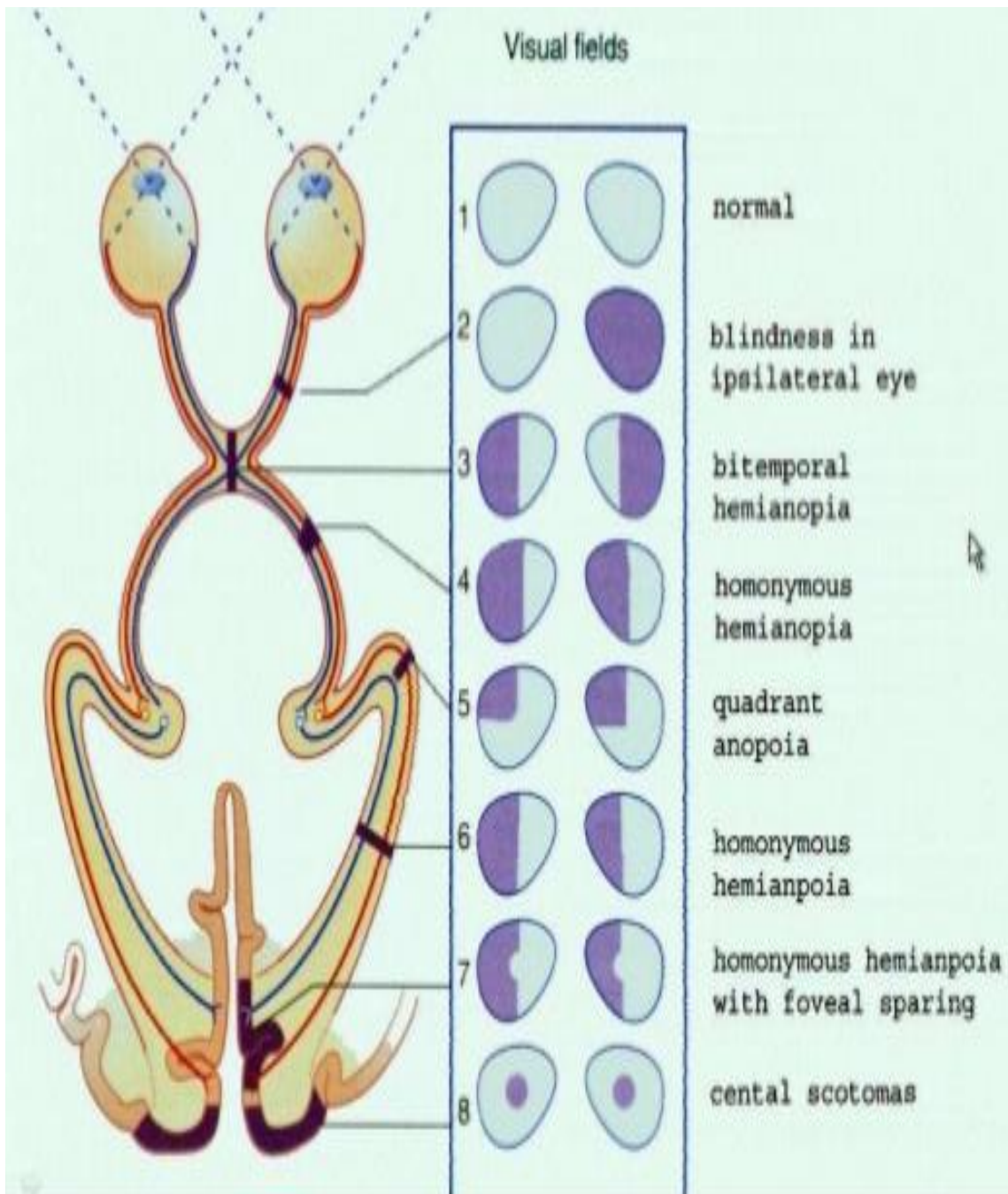


Central scotoma

- Affect central or macular vision



Central Scotoma

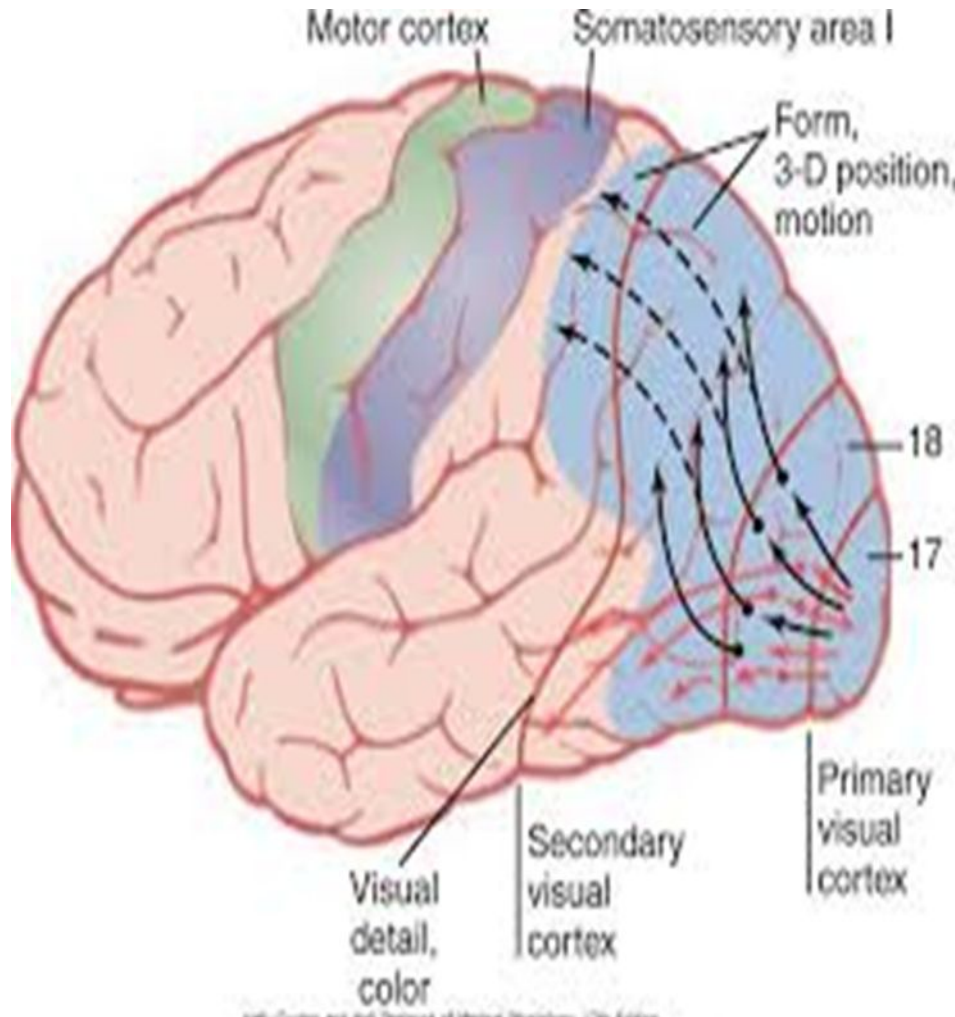


Lesions with visual defects

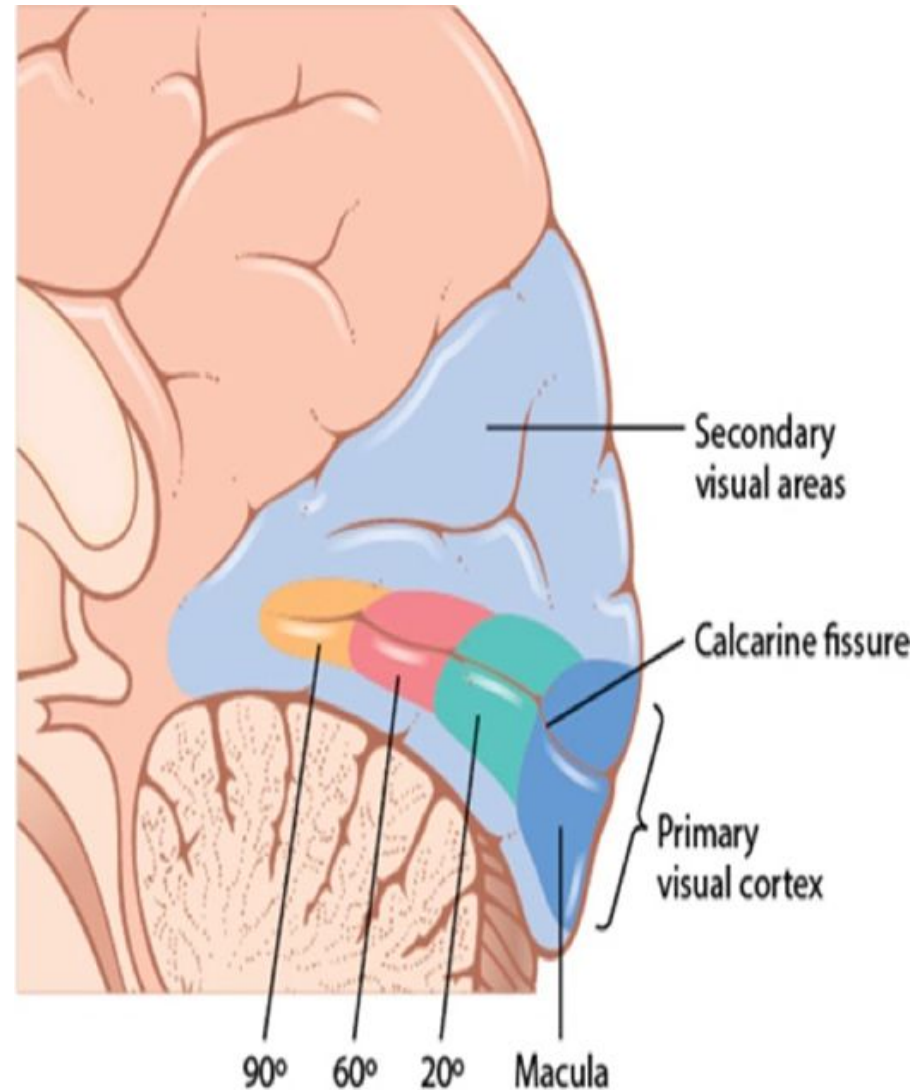
- Lesion of right optic nerve: Total blindness of right eye
- Lesion of optic chiasma: Bitemporal hemianopia
- Lesion of right optic tract: left homonymous hemianopia
- Lesion of outer fibers of right optic radiations: left upper quadrantanopia
- Lesion of right optic radiation: Left homonymous hemianopia.
- Lesion in right visual cortex: left homonymous hemianopia with macular sparing
- Macular damage :central vision lost

VISUAL CORTEX

- Visual cortex is located on the medial aspect of the occipital lobes and is divided into a primary and secondary visual cortex.
- The primary visual cortex (area 17) is called Striate cortex due to its striated appearance
- Lies in the calcarine fissure extending from the occipital pole on the medial aspect of each occipital cortex.
- This area is the terminus of direct visual signals from the eyes.



- Macula represents large area on Visual cortex receives stimuli from retinal fovea
- Signals from macula terminate near the occipital pole whereas signals from the more peripheral retina terminate anterior to the pole but still along the calcarine fissure.



- The secondary visual areas, also called *visual association areas*, lie lateral, anterior, superior, and inferior to the primary visual cortex.
- Secondary visual cortex (area 18) for analysis of visual meanings.

